New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: SIGNATURE:

ID NUMBER: EXAM ROOM: M A T H GROUP:

Final Exam

COURSE NAME: Game Theory COURSE NUMBER: ECON660

EXAMINATION DURATION: 120 MINUTES EXAM VERSION A

ADDITIONAL MATERIALS ALLOWED TO USE: Solve all 4 problems. Calculators or other materials are not allowed.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- Read and strictly follow extra instructions at the beginning of each Section.
- Put your signature and write your ID on even pages.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Question:	1	2	3
Points:	25	25	25
Score:			

Question:	4		Total
Points:	25		100
Score:			

SIGNATURE: ID:

1 Problems MOCK EXAM

Solve problems. Show your work.

1. (25 points) N firms are competing by Cournot. Demand is $P(Q) = A - BQ$, $TC(q) = F + CQ + (D/2)q^2$.

   - Solve for the individual quantities and prices.
   - What is the expected profit of each company?
   - What N sets the profit of each company to zero (disregard discreteness)?
   - What is the socially optimal N?

2. (25 points) This question is about the difference between Nash equilibrium, Bayesian Nash equilibrium, and Subgame Perfect Nash equilibrium.

   - Show a game with an example of NE that's not SPNE.
   - Show a game with an example of NE that's not BNE.

3. (25 points) In the principal-agent problem, explain why the principal, generically, is not taking away all the surplus of the agent.

4. (25 points) In a repeated game of Prisoner's Dilemma, both players with discount factor δ can try to collude: always play Don't Confess. It is clear that deviation is profitable; to discourage it, as soon as a deviation is detected, both players switch to Confess till the end of times.

   - Come up with payoffs of the Prisoner's Dilemma.
   - Calculate the δ such that both players are indifferent between maintaining collusion and deviating.